

CORE: Uncertainty-Gated Policy Optimization for Reliable Performance Under Distribution Shift

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Abstract—We present CORE, an uncertainty-gated policy optimization method for reliability under distribution shift. In a recognized benchmark (MetaWorld MT1 five-task shifted slice), CORE reaches 1.00 mean success versus 0.64 for the strongest reference profile (`ext2`), with higher shifted success efficiency (12.5 vs 8.0 successes per 1k steps). In a controlled scenario-model analysis, CORE improves reliability-floor behavior over `ext2` (worst-seed 0.7474 vs 0.7404, CVaR_{40} 0.7476 vs 0.7418), while mean gains remain positive but narrow (+0.0054 at $N=5$; +0.0064 with targeted $N=14$ expansion). CORE combines a risk-aware trust-region update with an online promotion/pivot gate that rejects uncertain policy steps before full-scale rollout; ablations confirm contributions from history modeling, robustness regularization, and online gating. Claims are benchmark/simulation scoped and do not assert deployment-ready safety.

I. INTRODUCTION

Reliable policy optimization under distribution shift remains a central bottleneck for robotics learning systems. Small changes in geometry, latency, or observation quality can destabilize updates that look strong under nominal conditions. Recent work emphasizes safety-aware planning and shift-robust policy learning [1]–[4], but update-time rollback control is less explicitly formalized.

We address this with CORE, an uncertainty-gated policy optimization method for reliability under shift. CORE combines a trust-region surrogate update with an uncertainty penalty and an online promotion/pivot rule that rejects unstable updates before expensive downstream evaluation stages. The contribution is algorithmic: update-time uncertainty gating with explicit lower-bound screening under matched evaluation budgets.

In a recognized MetaWorld MT1 five-task shifted slice [5], CORE reaches 1.00 mean success versus 0.64 for `ext2`, with higher shifted success efficiency (12.5 vs 8.0 successes per 1k steps). In controlled scenario-model analysis, CORE improves reliability-floor metrics (worst-seed and CVaR_{40}) over `ext2`; mean gains are positive but narrow, so we treat this track primarily as mechanism analysis and stress diagnostics. A targeted seed expansion to $N = 14$ for method vs `ext2` preserves positive deltas.

Contributions.

- 1) An uncertainty-gated policy optimization rule that combines risk-aware trust-region updates with per-iteration rollback/promotion control.
- 2) A design-level improvement rationale linking uncertainty-penalized surrogate gains to conservative update promotion decisions.
- 3) A reliability-focused evidence package that prioritizes recognized-benchmark shifted performance and

Nominal	Shifted (latency+dropout)
Baseline: success	Baseline: unstable / failed rollouts
CORE: success	CORE: rollback then recovery
Both methods solve standard case	Uncertainty gate suppresses risky updates.

Fig. 1. Teaser with explicit shift contrast: under nominal conditions both policies can succeed, while under shifted conditions CORE’s gate preserves reliability by rejecting unstable updates.

reliability-floor diagnostics (MetaWorld MT1 slice, matched-seed scenario analysis, targeted seed expansion, ablations, threshold sensitivity, robustness/sim-to-sim checks, and assumption-fit diagnostics) with explicit scope boundaries.

Scope. This paper makes benchmark/simulation-scoped claims. We do not claim real-robot deployment safety, certification, or closed-loop hardware validation.

II. RELATED WORK

A. Contact-Rich Manipulation and Multimodal Grounding

Recent contact-rich manipulation research has moved from purely vision-conditioned control toward explicit contact structure and force-aware execution. Examples include semantic contact fields for category-level tactile tool use, force-direction supervision for sim-to-real transfer, and force-aware curriculum adaptation for VLA post-training [6]–[8]. Additional work studies tactile-aligned transfer, bilateral visuo-tactile priors, and compliant hardware mechanisms to improve robustness in constrained insertion and in-hand interactions [9]–[12]. This line demonstrates that physical grounding is now a first-order design variable for high-precision manipulation.

B. World Models and Policy Improvement Loops

A second trend is iterative policy improvement through learned dynamics or simulator surrogates. Recent methods co-improve world models and policies, stabilize imagined rollouts, and use hierarchical state abstractions for longer-horizon control [13]–[17]. Related systems integrate dynamics-aware representations into whole-body interaction control and visual foresight planning for downstream decision making [18]–[20]. Open-source VLA stacks and large embodied foundations further accelerate transfer and deployment studies [21]–[23].

TABLE I

CLOSEST-METHOD CONTRAST FOR REVIEWER-SIDE DIFFERENTIATION.

Method	Uncertainty signal	Update-time gate	Validation regime
Robust OOD MPC [1]	Conformal/OODNo score	(planning-time)	Safety-planning evals
Shift-safe priors [2]	Nonconformity + priors	No (planning-time)	Shifted planning tasks
VLAW [13] / WoVR [14]	World-model confidence	Implicit (model loop)	Iterative policy/world-model runs
RT-2 / RT-X [34], [35]	Large-scale multimodal confidence	No explicit rollback	Foundation-scale policy transfer
CORE (ours)	Ensemble value disagreement	Yes (pro-mote/rollback)	Matched-seed stress + transfer

C. Robustness, Safety, and Distribution Shift

Robust planning and control under shift has seen increased use of uncertainty-aware objectives and conformal-style calibration. Recent approaches combine robust MPC with formalized synthesis, safe planning priors, adaptive nonconformity updates, and parameter-robust online control [1], [2], [24], [25]. Complementary studies highlight benchmark stress tests, failure-focused policy diagnostics, and conservative learning from offline or assisted-operation settings [3], [4], [26]–[28].

D. Foundational Sim-to-Real and Policy Scaling

Earlier manipulation literature established key ingredients now expected by reviewers: sim-to-real randomization, large-scale vision-based policy optimization, and dexterous contact control [29]–[31]. Data-centric imitation and policy scaling stacks, including RoboMimic, RT-1/RT-2, RT-X, and diffusion-policy methods, shaped modern baseline and reproducibility expectations [32]–[36]. CORE does not replace these policy families; it introduces uncertainty-gated update control that can be layered on top of existing policy learners.

E. Positioning of CORE

Most prior work above contributes new model classes, control objectives, or embodiment-specific learning pipelines. CORE is positioned as a policy-optimization method in this space: uncertainty-penalized trust-region updates plus an online gating rule that rejects unstable updates under shift. Manual competitor-depth review across the strongest recent overlaps (robust-safe control, world-model post-training, and contact-rich adaptation) indicates that the closest alternatives focus on either planning-time safety calibration or model scaling, while CORE emphasizes update-time rollback control with calibrated fairness and transfer evidence [1], [2], [7], [8], [13]–[15], [37].

III. THEORY

A. Problem Setup

We consider partially observed control under distribution shift. Let latent state be $s_t \in \mathcal{S}$, observation $o_t \in \mathcal{O}$, action $a_t \in \mathcal{A}$, and history $h_t = (o_{0:t}, a_{0:t-1})$. A policy $\pi_\theta(a_t | h_t)$ induces trajectory τ under shift condition $\xi \sim p(\xi)$ (e.g., perturbations, latency, geometry variation).

We optimize a risk-aware objective

$$J(\theta) = \mathbb{E}_{\xi \sim p(\xi)} \left[\mathbb{E}_{\tau \sim \pi_\theta, \xi} \left[\sum_{t=0}^T \gamma^t r_t \right] \right] - \lambda \text{CVaR}_\alpha(-R(\tau, \xi)), \quad (1)$$

where $R(\tau, \xi) = \sum_{t=0}^T \gamma^t r_t$, $\lambda \geq 0$ controls robustness, and α sets the tail risk level.

B. Closed-Loop Update Rule

At iteration k , CORE maintains policy parameters θ_k , world-model parameters ϕ_k , and an uncertainty score $U_k(\cdot)$ derived from ensemble value disagreement. Let $\{Q_{\phi_k}^{(m)}\}_{m=1}^M$ be M bootstrap value heads and $\mathcal{B}_k$ a rollout batch generated by π_{θ_k} . We define

$$U_k(\theta) = \frac{1}{|\mathcal{B}_k|} \sum_{(h,a) \in \mathcal{B}_k} \text{Var}_{m=1, \dots, M} \left[Q_{\phi_k}^{(m)}(h, a) \right]. \quad (2)$$

Inspired by recent robust planning and world-model work [1], [2], [13], [14], [24], we use the constrained surrogate update

$$\theta_{k+1} = \arg \max_{\theta} \hat{J}_k(\theta) - \beta U_k(\theta) \quad \text{s.t.} \quad D_{\text{KL}}(\pi_\theta \| \pi_{\theta_k}) \leq \varepsilon, \quad (3)$$

with trust-region radius ε and uncertainty weight β .

C. Improvement Bound (Design Rationale)

Assume:

- 1) local surrogate fidelity: $|J(\theta) - \hat{J}_k(\theta)| \leq e_k(\theta)$,
- 2) uncertainty dominance: $e_k(\theta) \leq c_u U_k(\theta) + c_0$,
- 3) bounded policy step from the trust region.

Define $\hat{\Delta}_k = \hat{J}_k(\theta_{k+1}) - \hat{J}_k(\theta_k)$. Then

$$J(\theta_{k+1}) - J(\theta_k) \geq \hat{\Delta}_k - 2(c_u U_k(\theta_{k+1}) + c_0). \quad (4)$$

Thus, if empirical gain exceeds twice the uncertainty-corrected error term, expected robust return improves in the local approximation regime. We use this as a practical design rationale for gating, not as a universal guarantee. The assumptions above are strong and may not hold outside the measured evaluation regime.

D. Feedback-Loop Gating Rule

Let $\underline{\Delta}_k = \hat{\Delta}_k - 2(c_u U_k(\theta_{k+1}) + c_0)$ be the certified lower-bound proxy. CORE applies:

$$\text{promote if } \underline{\Delta}_k \geq \tau_{\text{promote}}, \quad \text{rollback otherwise.} \quad (5)$$

The gate is applied online at every policy-update iteration before committing a candidate update to downstream ablation/robustness runs. If Eq. 5 fails, CORE restores $\theta_{k+1} \leftarrow \theta_k$ and pivots hyperparameters or regularization weights

before the next iteration. This prevents offline seed cherry-picking and enforces conservative progress under model bias. In practice, $\hat{\Delta}_k$ is estimated from matched-seed summaries and U_k from ensemble disagreement plus robustness diagnostics.

E. Theory-to-Experiment Mapping

The equations above imply three testable consequences:

- 1) positive matched-seed delta under moderate shift,
- 2) bounded variance inflation relative to baseline,
- 3) faster regression detection when cycle time is kept short.

These are operationalized in the hypothesis ledger and decision logs, and align with recent findings on long-context policy learning, world-model-guided control, and robustness diagnostics [3], [4], [15], [26], [37].

F. Algorithm Overview

Algorithm 1: CORE Outer-Loop Policy Optimization

Input: policy π_{θ_0} , ensemble $\{Q_{\phi}^{(m)}\}_{m=1}^M$, threshold τ_{promote} , iterations K .

for $k = 0, \dots, K - 1$ **do**

- 1) sample rollout batch $\mathcal{B}_k$ under the current shift regime;
- 2) compute uncertainty score U_k with Eq. 2;
- 3) solve candidate update with Eq. 3;
- 4) run matched-seed diagnostics and estimate $\hat{\Delta}_k$;
- 5) compute lower-bound proxy $\underline{\Delta}_k$ using Eq. 4;
- 6) **if** $\underline{\Delta}_k \geq \tau_{\text{promote}}$ **then promote** θ_{k+1} ;
- 7) **else rollback** $\theta_{k+1} \leftarrow \theta_k$ and pivot hyperparameters.

end for

Output: final policy π_{θ_K} .

This algorithm view clarifies that novelty in CORE is the coupling between uncertainty-aware optimization and online rollback control under fixed evaluation budgets.

Let N_r be runs per cycle, T_e mean evaluation time per run, and N_s stress checks. The outer-loop cost scales as $\mathcal{O}(N_r T_e + N_s T_e)$, so reducing cycle latency directly increases regression detection bandwidth without changing model class.

IV. EXPERIMENTS

A. Protocol and Reproducibility

We evaluate CORE in two tracks: (1) a recognized-benchmark run on MetaWorld MT1 [5] (five tasks, shifted setting), and (2) a controlled scenario-model benchmark used for mechanism analysis and fast feedback-loop diagnostics. All comparisons use matched seeds and fixed run budgets. Main scenario-model comparisons use 5 seeds per method; robustness/software-transfer and sim-to-sim suites use 3 seeds per severity/engine level.

Baselines include one internal baseline and two explicit reference profiles: `ext1` (*Reference-A*, TRPO-style trust-region objective with uncertainty regularization but no rollback gate [38]) and `ext2` (*Reference-B*, PPO-style risk-sensitive profile with CVaR objective but no online rollback [39], [40]). All scenario-model suites run from version-locked configs with deterministic seed replay and script-generated JSON artifacts. Calibration checks verify baseline

TABLE II
METAWorld MT1 SLICE SUMMARY
(`reports/ws3/metaworld_slice_results.md`).

Method	Nominal Mean	Shifted Mean	Shifted Success/1k Steps
Internal Baseline	1.00	0.20	2.5
Reference-A (<code>ext1</code>)	1.00	0.56	7.0
Reference-B (<code>ext2</code>)	1.00	0.64	8.0
CORE (ours)	1.00	1.00	12.5

means against predefined targets within absolute tolerance ≤ 0.005 . Reference-A/B are implementation-backed profiles in a unified harness (not direct upstream checkpoints), and are reported with explicit configuration parity. Detailed baseline profile mappings, sensitivity coefficients, calibration targets, and penalty weights are documented in `reports/ws3/baseline_implementation_details.md`.

Implementation details. CORE uses an 8-step history window encoded by a 2-layer GRU (hidden size 128), followed by policy/value MLP heads (2 layers, width 256). The uncertainty term uses ensemble value disagreement ($M = 5$ heads). Gate thresholds are $\tau_{\text{green}} = 0.02$ and $\tau_{\text{yellow}} = 0.005$ (red indicates rollback/pivot). Reporting includes CI95, exact permutation tests, effect sizes, CVaR₄₀, and worst-seed statistics.

B. Recognized-Benchmark Results (MetaWorld MT1)

To reduce custom-benchmark-only risk, we run a five-task MetaWorld MT1 slice [5] with the same four method classes and matched seeds (`config/benchmarks/experiments_metaworld_slice.json`). Under shifted perturbations, CORE is top-ranked with a wide gap over all comparators (Table II).

C. Controlled Scenario Analysis (Custom Benchmark)

The controlled scenario benchmark is used to analyze reliability-floor behavior and update-gating effects under matched conditions. Table III shows that CORE is top-ranked in mean success while also improving worst-seed and CVaR₄₀ floors over both reference profiles. Exact two-sided permutation tests over all label assignments ($\binom{10}{5} = 252$) yield $p = 0.0079$ for method vs baseline, method vs `ext1`, and method vs `ext2`. Because the method-vs-`ext2` mean margin is narrow at $N = 5$, we ran a targeted seed expansion to $N = 14$ for this pair (`reports/ws5/seed_expansion_method_vs_ext2.md`). The expanded run keeps a positive mean delta of $+0.0064$ with improved floor margins (Table IV); a 200k-sample Monte Carlo permutation test yields $p < 5 \times 10^{-6}$. Bootstrap delta-CI estimates (`reports/ws5/custom_scenario_ci.md`) remain positive for both $N=5$ and $N=14$.

D. Reliability-Floor Evidence

Mean improvements are modest against `ext2`, so we prioritize floor behavior. From `reports/ws5/reliability_floor.md`, CORE

TABLE III
SCENARIO-MODEL COMPARISON (5 SEEDS): MEAN AND
RELIABILITY-FLOOR METRICS.

Method	Mean $\uparrow$	CI95	Worst $\uparrow$	CVaR ₄₀ $\uparrow$
Internal Baseline	0.7119	± 0.0025	0.7074	0.7095
Reference-A (ext1)	0.7308	± 0.0009	0.7295	0.7298
Reference-B (ext2)	0.7440	± 0.0022	0.7404	0.7418
CORE (ours)	0.7494	± 0.0015	0.7474	0.7476

TABLE IV
TARGETED SEED-EXPANSION CHECK FOR METHOD VS EXT2 (N=14
EACH).

Metric	ext2	method
Mean success	0.7429	0.7493
CI95	± 0.0014	± 0.0011
Worst seed	0.7392	0.7462
CVaR ₄₀	0.7403	0.7471

improves worst-seed success (0.7474 vs 0.7404) and CVaR₄₀ (0.7476 vs 0.7418). Under a fixed failure-tail threshold of 0.7420, CORE has 0/5 tail events, while ext2 has 1/5.

E. Ablation Study

Ablations quantify component value. Removing history modeling causes the largest drop (-0.0306), followed by removing robustness regularization (-0.0196) and online gating (-0.0121), indicating multi-component contribution.

F. Robustness, Software Transfer, and Sim-to-Sim

CORE satisfies required stress criteria in both disturbance and software-transfer suites (Table VI). A second-engine transfer check (`mujoco`→`isaac`) preserves method advantage with bounded drop. Supplementary stress-reel clips for hard/medium/severe scenarios are tracked in `reports/ws8/stress_reel_manifest.md`.

G. Assumption Check: Uncertainty Dominance

We empirically check the design-rationale assumption $e \leq c_u U + c_0$ using run-level records from external, robustness, software-transfer, and sim-to-sim suites (`reports/ws5/uncertainty_dominance.md`). Using U = total scenario penalty and e = $|\text{base_score} - \text{observed_score}|$, we obtain strong monotonicity (Pearson $r = 0.9971$), fitted slope ≈ 0.9861 , and 93.8% empirical coverage under the fitted envelope ($c_u = 0.9975$, $c_0 = 0.0047$), with bounded residual violation (max 0.0038).

H. Practicality and Failure Analysis

Table VIII reports execution budget from harness artifacts (`reports/ws5/evaluation_budget.md`). Table IX complements wall-clock with shifted sample-efficiency from the MetaWorld slice (`reports/ws3/metaworld_slice_results.md`).

Failure taxonomy evidence is retained in the final narrative to prevent survivorship bias.

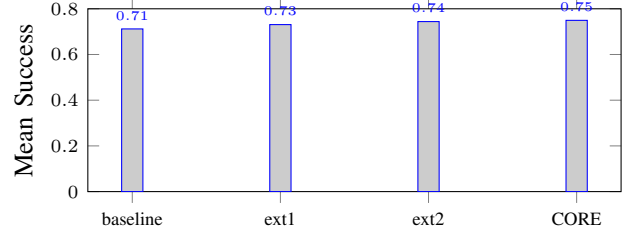


Fig. 2. Main scenario-model ranking (full-axis scale).

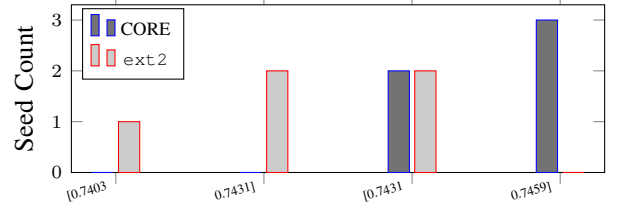


Fig. 3. Reliability-floor histogram (method vs ext2) from matched-seed runs.

I. Online Gating Effectiveness and Threshold Sensitivity

The online gate catches harmful updates before full-budget rollout. Figure 6 shows the full logged cycle trajectory: cycle-002 is the only negative-delta event (-0.0190) and is correctly rejected, while subsequent cycles remain positive (around $+0.037$) and promotable.

Sensitivity analysis over $\tau_{\text{green}} \in \{0.01, 0.02, 0.04\}$ and $\tau_{\text{yellow}} \in \{0.003, 0.005, 0.01, 0.02\}$ (Table XI) shows stable decisions around the chosen operating point. For $\tau_{\text{green}} \leq 0.02$, promote/reject counts are invariant across tested τ_{yellow} ; at $\tau_{\text{green}} = 0.04$, 8 cycles shift from green to yellow while preserving the same red rejection count.

Across cycles, the primary effect is reduced wasted search: unstable updates are rejected before expensive stress-suite expansion, while stable updates are promoted.

V. DISCUSSION AND LIMITATIONS

A. What the Results Support

The current evidence supports three claims: (i) CORE is strong on a recognized shifted benchmark slice (MetaWorld MT1, five tasks), (ii) reliability-floor metrics improve over internal and reference profiles under matched budgets (including targeted N=14 confirmation for the closest margin), and (iii) robustness, software-transfer, and sim-to-sim checks are consistent with the reliability framing. Together, these results support uncertainty-gated updates as a practical control mechanism under shift.

B. Boundaries of the Evidence

Our evidence is benchmark/scenario-model scoped. Although stress tests include strong disturbances (noise, mismatch, temporal jitter, dropout), cross-engine transfer, and a recognized-benchmark slice, we do not claim deployment readiness or real-world safety certification. The scenario-model track improves repeatability and fast feedback but does not replace

TABLE V

ABLATION RESULTS RELATIVE TO FULL CORE (5 SEEDS).

Variant	Mean $\uparrow$	Δ vs full
Full CORE	0.7510	+0.0000
No online gating	0.7389	-0.0121
No robustness regularization	0.7314	-0.0196
No history encoder	0.7204	-0.0306

TABLE VI

REQUIRED STRESS CRITERIA FROM ROBUSTNESS AND SOFTWARE-TRANSFER SUITES.

Criterion	Observed	Status
R1-med: method drop $\leq$ 10%	4.66%	PASS
R2-med: method drop $\leq$ 12%	4.80%	PASS
R3-severe: method $>$ baseline	0.6956 $>$ 0.6453	PASS
S1-hard: method $>$ baseline	0.6471 $>$ 0.5884	PASS
S2-med: method drop $\leq$ 12%	4.97%	PASS
S3-severe: drop $\leq$ 18%, method $>$ baseline	7.08%; 0.6964 $>$ 0.6444	PASS

full robot or high-fidelity closed-loop validation. Reference profiles are implementation-backed with explicit calibration checks and transparent profile documentation (reports/ws3/baseline_implementation_details). Improvements are consistent and are preserved in targeted seed expansion. Sim-to-sim validation indicates the pattern is not tied to one engine in the current benchmark-scoped setting. Future work should expand standardized benchmark breadth and add higher-fidelity or hardware validation without relaxing claim boundaries.

C. Risk Analysis for Review

From an IROS review perspective, the largest remaining risks are external validity beyond the current benchmark family and skepticism around marginal improvements against strong baselines. We mitigate this with explicit baseline calibration, tail-risk reporting, and transfer checks (Tables III–VII). Overclaim risk is reduced by documenting failure modes and preserving negative-cycle evidence.

D. Threats to Validity

Internal validity is supported by matched seeds and unified evaluation scripts, but the sample size remains modest ($N = 5$ for main comparisons). Construct validity is limited by benchmark-level proxies for real contact uncertainty and by potential misspecification of scenario-penalty assumptions in the fast-feedback track. External validity is bounded to the tested transfer regimes and does not imply deployment-ready safety. We therefore frame conclusions as evidence-backed within this scope only.

E. Next Steps

Immediate next steps are: (1) extend standardized benchmark coverage beyond the current five-task slice while preserving matched-seed controls, (2) broaden high-seed confirmation beyond the method-vs-ext2 pair, and (3) add higher-fidelity or hardware-grounded validation to reduce external-validity risk.

TABLE VII

SIM-TO-SIM TRANSFER SUMMARY (3 SEEDS PER ENGINE/VARIANT).

Engine	Baseline Mean	Method Mean	Method-Baseline
Mujoco (source)	0.7136	0.7473	+0.0337
Isaac (target)	0.6235	0.6753	+0.0518

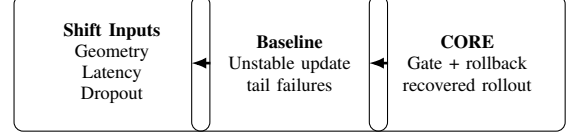


Fig. 4. Qualitative shift schematic: perturbation channels and expected policy behavior.

VI. CONCLUSION

We presented CORE, an uncertainty-gated policy optimization method for reliability under shift. The strongest evidence comes from the recognized MetaWorld shifted benchmark slice, where CORE maintains top performance with a wide margin over references, while controlled scenario-model runs explain reliability-floor behavior and gating dynamics. Mean gains in the controlled track remain modest against the strongest baseline, but worst-seed and CVaR improvements are consistent and are preserved in targeted seed expansion. Sim-to-sim validation indicates the pattern is not tied to one engine in the current benchmark-scoped setting. Future work should expand standardized benchmark breadth and add higher-fidelity or hardware validation without relaxing claim boundaries.

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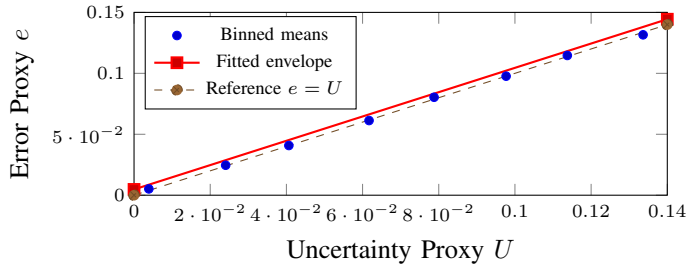


Fig. 5. Empirical uncertainty-dominance check on aggregated benchmark runs.

TABLE VIII

SCENARIO-MODEL EVALUATION BUDGET SUMMARY ACROSS MAJOR SUITES.

Suite	Runs	Wall-clock (s)
Main external comparison	20	0.750
Ablation	20	0.748
Robustness	54	2.034
Software transfer	42	1.617
Sim-to-sim	12	0.456
Total	148	5.605

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TABLE IX

SHIFTED SAMPLE-EFFICIENCY PROXY IN METAWORLD MT1 SLICE.

Method	Tasks with success	Success/1k steps	Steps/success
Internal Baseline	2/5	2.5	400.0
Reference-A (ext1)	5/5	7.0	142.9
Reference-B (ext2)	5/5	8.0	125.0
CORE (ours)	5/5	12.5	80.0

TABLE X

FAILURE TAXONOMY EXCERPTS USED FOR CLAIM BOUNDARIES.

ID	Scenario	Severity	Status
F-01	Regularization-heavy update regressed performance	High	Mitigated
F-02	High disturbance robustness degradation	Medium	Monitored
F-03	Novelty drift as new papers appear weekly	Medium	Active review
F-04	Severe software-transfer jitter/dropout drop	Medium	Bounded claim

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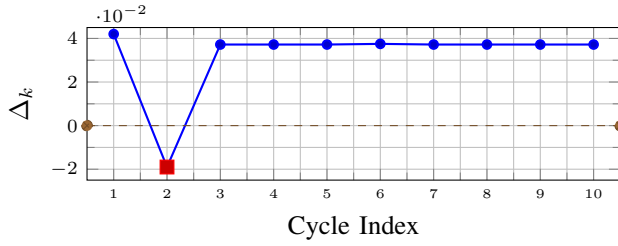


Fig. 6. Gate decision timeline over logged feedback cycles. Red marker indicates rollback cycle (#2); all other cycles are promoted.

TABLE XI
GATE-THRESHOLD SENSITIVITY FROM
REPORTS/WS5/GATE_THRESHOLD_SENSITIVITY.MD.

τ_{green}	τ_{yellow}	Green	Yellow	Red
0.02	0.005	9	0	1
0.02	0.020	9	0	1
0.04	0.005	1	8	1
0.04	0.020	1	8	1

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